GRANISETRON

Long-acting
Patch available

CHEMICAL NAME 1-methyl-N-[(1R,3r,5S)-9-methyl-9-azabicyclo[3.3.1]nonan-3-yl]-1H-indazole-3-carboxamide (hydrochloride)

CLASS 5-HT3 receptor antagonist antiemetic

TRADE NAMES Kytril · Sancuso (patch) · Granitron

THERAPEUTIC USES / INDICATIONS

- Chemotherapy-induced nausea and vomiting (acute and delayed).
- Radiotherapy-induced nausea and vomiting.
- Post-operative nausea and vomiting.

PHARMACODYNAMICS

- Highly selective **5-HT3** antagonist peripherally (vagal) and centrally (CTZ).

ADMINISTRATION

- **IV** bolus over 30 s or infusion over 5 min.
- **PO** tablets 1 h before chemotherapy.
- **Transdermal patch** applied 24–48 h before chemotherapy to upper arm; worn up to 7 days.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation, headache.
- Asthenia, somnolence.
- Patch-site reactions.

ADVERSE EFFECTS (SERIOUS)

- QT prolongation (less than ondansetron).
- Serotonin syndrome.
- Hypersensitivity.

DOSAGES

- **Chemotherapy IV:** 1–3 mg (10–40 mcg/kg) before treatment; max 9 mg/24 h.
- **PO:** 1–2 mg 1 h before chemotherapy (2 mg once or 1 mg twice daily).
- **Patch:** 3.1 mg/24 h (one patch).
- **PONV:** 1 mg IV.

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to 5-HT3 antagonists.
- **Caution:** subacute bowel obstruction, QT risk.

KEY INTERACTIONS

- **Serotonergic drugs:** serotonin syndrome.
- **QT-prolonging drugs:** caution.

PHARMACOKINETICS

- **Onset:** IV minutes. **Duration:** up to 24 h.
- Metabolised in liver (CYP3A4).
- Excreted in urine and faeces. **t½:** 9 h (longer in cancer patients).

NURSING MANAGEMENT

- Time doses before chemotherapy.
- Patch: apply to clean, intact skin; protect from sunlight/heat lamps.
- Monitor bowel function and vomiting episodes.

PATIENT EDUCATION

- Cover the patch from direct sunlight.
- Report constipation lasting >3 days.
- Take tablets 1 h before chemotherapy.

KEY POINTS

- 5-HT3 antagonist, longer acting than ondansetron.
- Constipation, headache.
- Once-daily dosing or 7-day patch.
- Chemotherapy nausea.